From candidate to a discriminating experiment

PROPOSED comparisons · no pathway mediation, secretion or repair function established by this screen

Observed RNA	Working question	Decisive first comparison	Alternative / informative failure
Tigit and Prdm1 decrease. Areg interaction is fragile.	Does Ezh2 loss impair inducible TIGIT–Blimp-1 output in viable activated Tregs?	Matched HET/KO activation, with versus without TIGIT engagement: surface TIGIT, Blimp-1 and secreted AREG per viable cell.	Equal output despite RNA changes weakens this route. A general activation or survival defect can mimic it. Repair needs its own functional assay.
Ccr2 decreases, together with several other chemokine receptors.	Does reduced CCR2 contribute to altered localization, cell fitness, or simply mark a changed Treg state?	Measure CCR2 protein, direct chemotaxis, CD25/IL-2 response and viable recovery as separate endpoints.	Abundance is not migration. Migration preserved with altered fitness favors a different explanation; shared-state shifts can explain both.
Igfbp4 increases. External H3K27me3 context is compatible but unmatched.	Is IGFBP4 a functional growth-factor buffer, a compensatory response, or a population marker?	First confirm Treg-source protein. Then perturb IGFBP4 under defined IGF exposure and measure signaling, survival and suppression separately.	Absent protein or unchanged function weakens the mediator model. Opposite effects support compensation; IGF effects depend on cell state.

Premises: TIGIT study (2025), CCR2 study (2018) and IGF study (2020); exact locators in TARGET_SCREEN_RESULT.md.
Unit: independently obtained mouse/preparation; wells are technical. Effect margin, sample size and laboratory feasibility remain unspecified.